WEEK 9: HOUSING DATA CLEANING & TRANSFORMATION

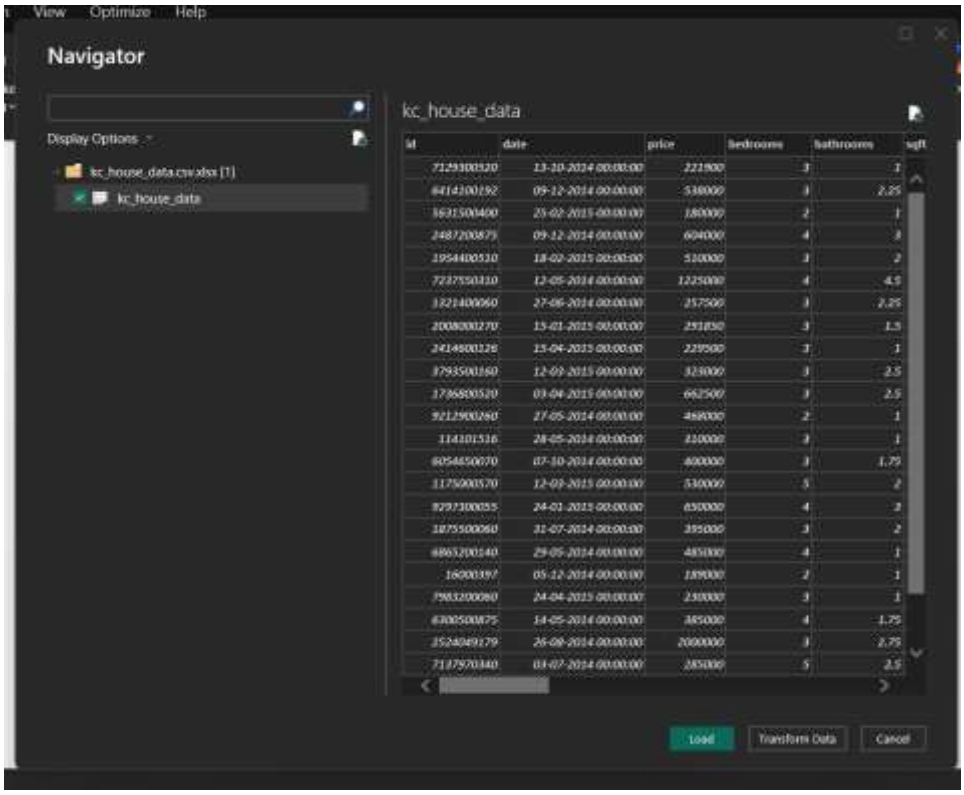
Aim:

To clean, transform, and prepare the KC House Sales dataset using Power BI and Power Query.

Procedure

Step 1: Import the Dataset:

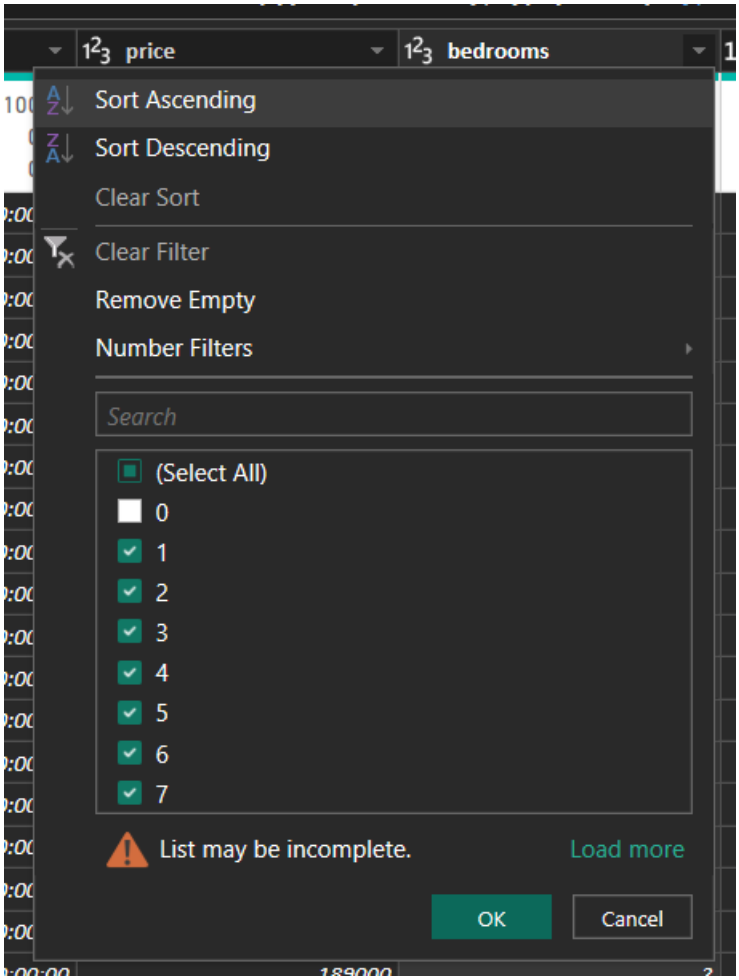
1. Open Power BI Desktop.
2. Click **Home** → **Get Data** → **Text/CSV**.
3. Select the kc_house_data.csv file and click **Open**.
4. In the preview window, click **Transform Data** to open Power Query Editor.



Step 2: Check Data Types and Clean the Data:

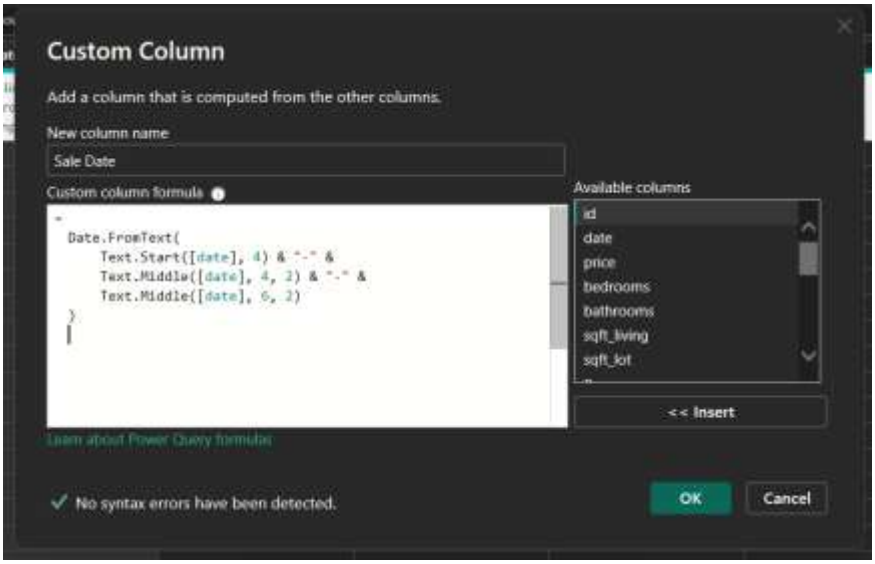
Set columns such as price, bedrooms, bathrooms, sqft_living, sqft_lot, yr_built, and yr_renovated to suitable numeric data types.

Select the bedrooms column, click its filter arrow, uncheck 0, and click **OK**.



Step 3: Create the SaleDate Column:

1. Select the original date column.
2. Click the **Add Column** tab.
3. Select **Date** → **Date Only**.
4. Rename the newly created column to SaleDate.
5. Check that its data type is **Date**.



Step 3: Create the Sale Year Column:

1. Select the SaleDate column.
2. Click **Add Column** → **Date** → **Year** → **Year**.
3. Rename the new column to Sale Year.
4. Set the data type to **Whole Number**.

Step 4: Create the Sale Month Column:

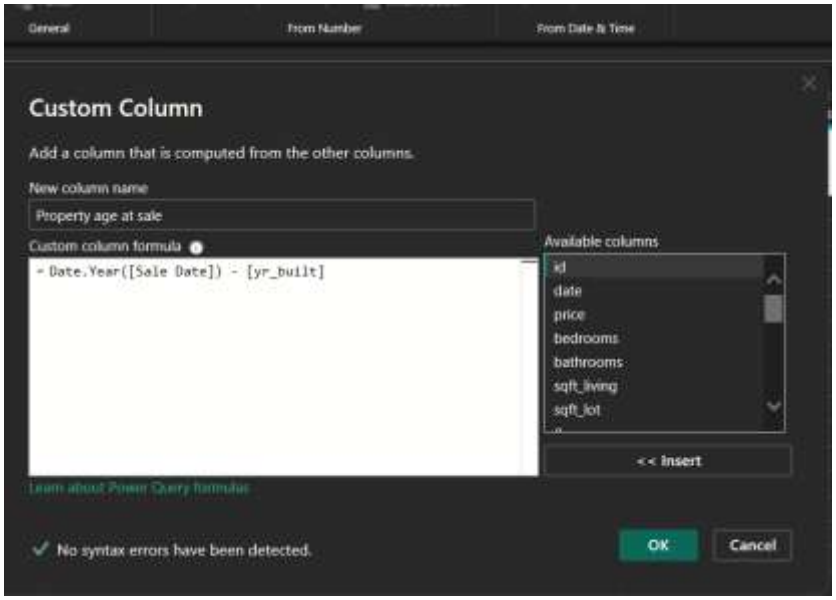
1. Select the SaleDate column.
2. Click **Add Column** → **Date** → **Month** → **Month**.
3. Rename the new column to Sale Month.
4. Set the data type to **Whole Number**.

Step 5: Create the Property Age at Sale Column:

1. Click **Add Column** → **Custom Column**.
2. Enter Property Age at Sale as the new column name.
3. Enter the following formula:

   Date.Year([SaleDate]) - [yr_built]

4. Click **OK** and set the data type to **Whole Number**.



Step 6: Create the Price per SqFt Column

1. Click **Add Column** → **Custom Column**.
2. Name the column Price per SqFt.
3. Enter the formula:

   if [sqft_living] = null or [sqft_living] <= 0 then null else [price] / [sqft_living]

4. Click **OK** and set the data type to **Decimal Number**.

Step 7: Create the Total Area Column

1. Click **Add Column** → **Custom Column**.
2. Name the column Total Area.
3. Enter the formula:

   if [sqft_living] = null or [sqft_lot] = null then null else [sqft_living] + [sqft_lot]

4. Click **OK** and set the data type to **Whole Number**.

Step 8: Create the Is Renovated Column

1. Click **Add Column** → **Custom Column**.
2. Name the column Is Renovated.
3. Enter the formula:

   if [yr_renovated] = null then null else if [yr_renovated] > 0 then "Yes" else "No"

4. Click **OK** and set the data type to **Text**.

Step 9: Categorize Property Prices

1. Click **Add Column** → **Custom Column**.
2. Name the column Price Tier.
3. Enter the formula:

if [price] = null then null else if [price] < 300000 then "Budget" else if [price] < 600000 then "Mid-Range" else if [price] < 1000000 then "Premium" else "Luxury"

4. Click **OK** and set the data type to **Text**.

Step 10: Load the Cleaned Data

1. Click **Home** → **Close & Apply**.
2. Wait for Power BI to load the transformed data.
3. Save the project .

Result

The King County House Sales dataset was cleaned and transformed using Power Query. The required date attributes, derived parameters, property size categories, and price tiers were created, and the cleaned data was loaded into Power BI.